INNOPOLIS UNIVERSITY

SNE MASTER TRACK

2025

COURSE: [F25] Networks Engineering

INR LAB 3 Report

On

STP, VLAN, and Fault Tolerance

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1. Preparation:

a. Read the notes at the end.

b. Select a virtual switching solution that you would like to try. For example (Mikrotik, vyos, Cumulus VX).

c. GNS3 already have a template for these switches (Mikrotik, vyos, Cumulus VX), try to use these templates as it will save you a lot of time and troubleshooting.

d. Be sure to take a snapshot of your infrastructure before you start

2. VLANs:

a. Change the topology of your network to as follows, and make the necessary configurations.

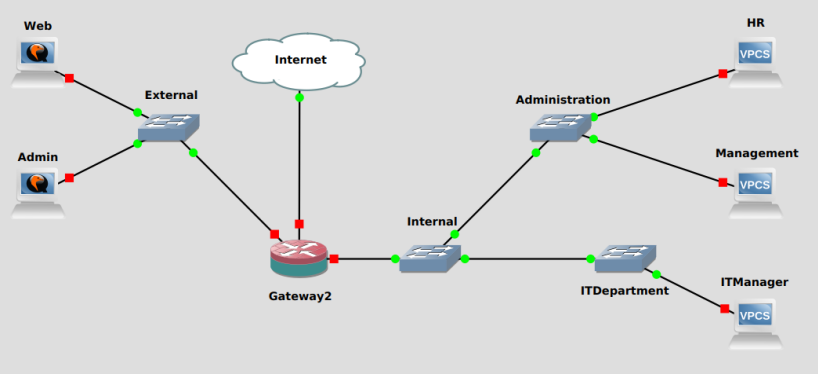


Figure-1

b. Configure the switches and make sure you have connectivity between the hosts.

c. How do VLANs work at a packet level? What are the two main protocols used for this?

d. What is the Native VLAN?

e. Configure the VLANs on the switches to isolate the two virtual networks as follow.

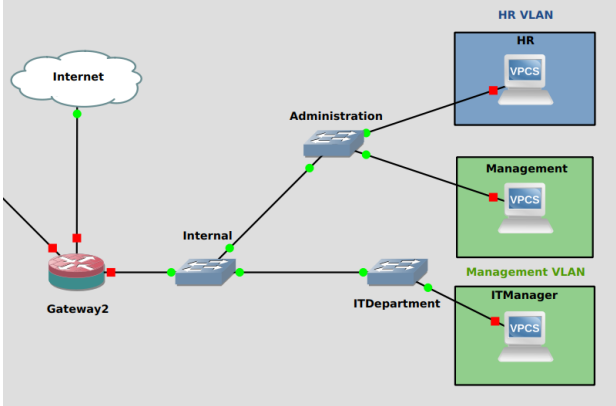


Figure-2

f. Ping between ITManager and HR, do you have replies? Ping between ITManager and Management, do you have replies? Can you see the VLAN ID in Wireshark?

g. Configure Inter-VLAN Routing between Management VLAN and HR VLAN and Show that you can now ping between them.

3. Fault Tolerance:

a. What is Link Aggregation? How does it work (briefly)? What are the possible configuration modes?

b. Use link aggregation between the Web and the Gateway to have Load Balancing and Fault Tolerance as follows.

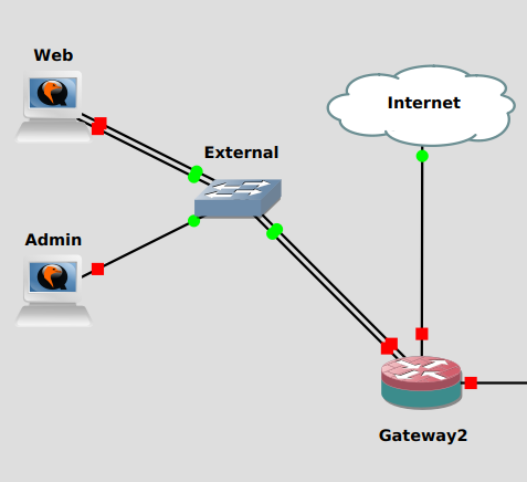


Figure-3

c. Test the Fault Tolerance by stopping one of the cables and see if you have any downtime.

4. STP (bonus):

a. Change the topology as follows and Disable STP on the Internal network switches.

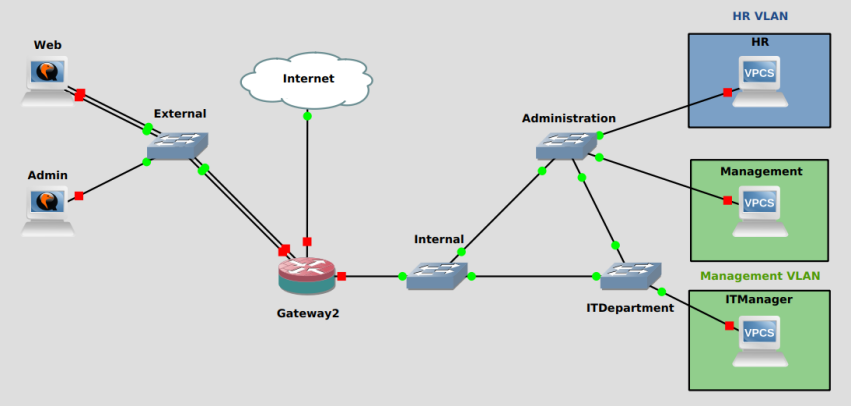


Figure-4

b. Send a broadcast ping request to the PCs connected to the Internal Network.

c. What can you notice? Why did this happen? What are the implications of this on the network?

d. Enable back STP on the Switches and do the experiment again.

e. Can you see STP traffic? Explain it briefly.

f. Configure the switches to have the *Internal* as the Root switch.

g. Would we need STP between routers?

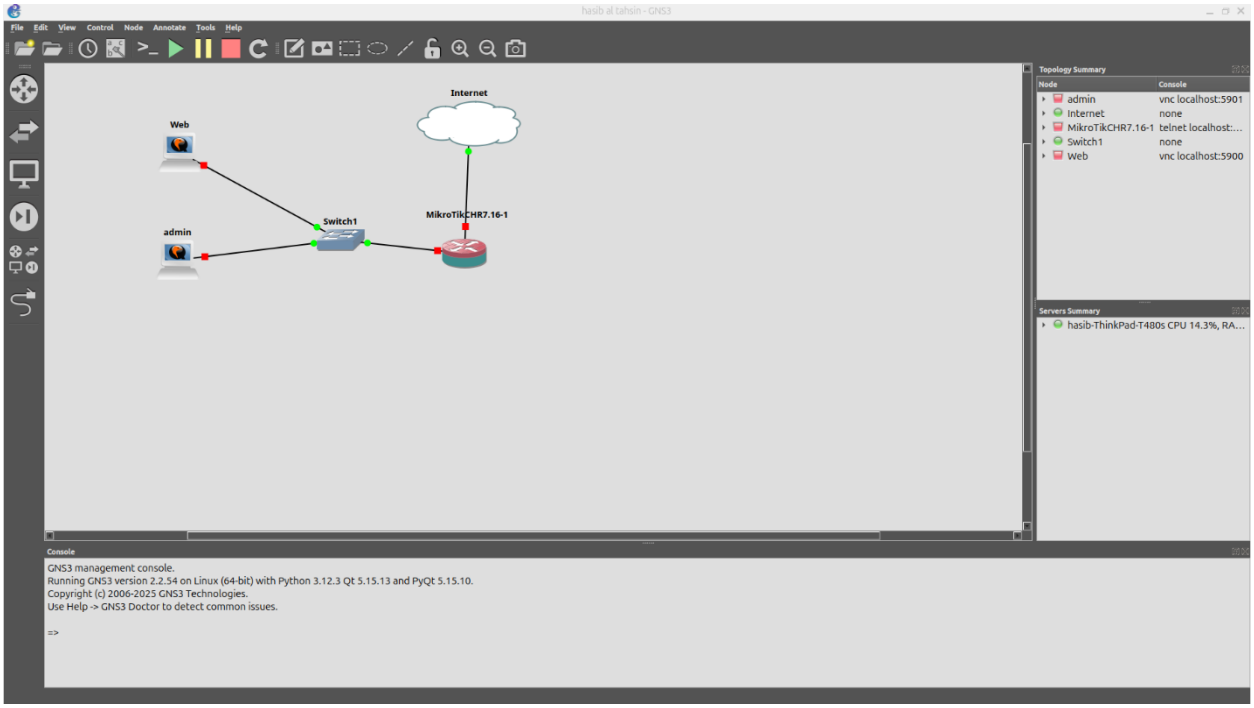
Notes:

- Make the report as technical as possible (no installation guide please).
- Try to include a network scheme in your report
- If you paste some data (routing table), please make sure it is readable and the format did not change
- If you want to include a command in the report, please highlight it (bold, italic, different format, ...)
- Keep your lab project file, you might use it for the next lab.
- For STP, you will have a CPU load of 100%, please be careful

Solutions:

Task 1 – Preparation:

a. I have read the end notes very carefully and tried my best to follow till the end.
b. I have chosen Mikrotik's switching solution among the suggested kora appliance for virtual switching solution, which is also convenient and relatively easy to work with.
c. GNS3 has switches templates, out of which I have chosen Mikrotik template.
d. I have added the snapshot before starting the infrastructure design of this lab, which I have added below.



Task 2 - VLANs:

a. Changing the Topology (Figure-1) and Basic Configuration

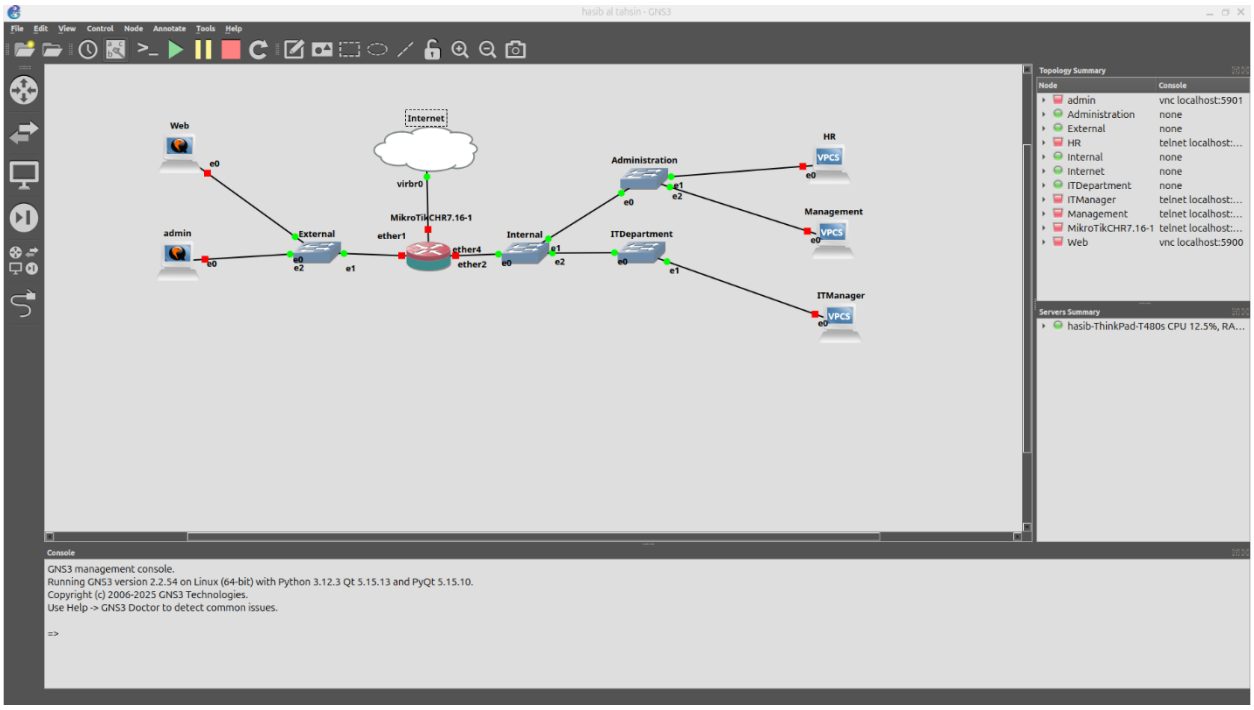
I first did setup the network topology as per figure below.

Devices Used:

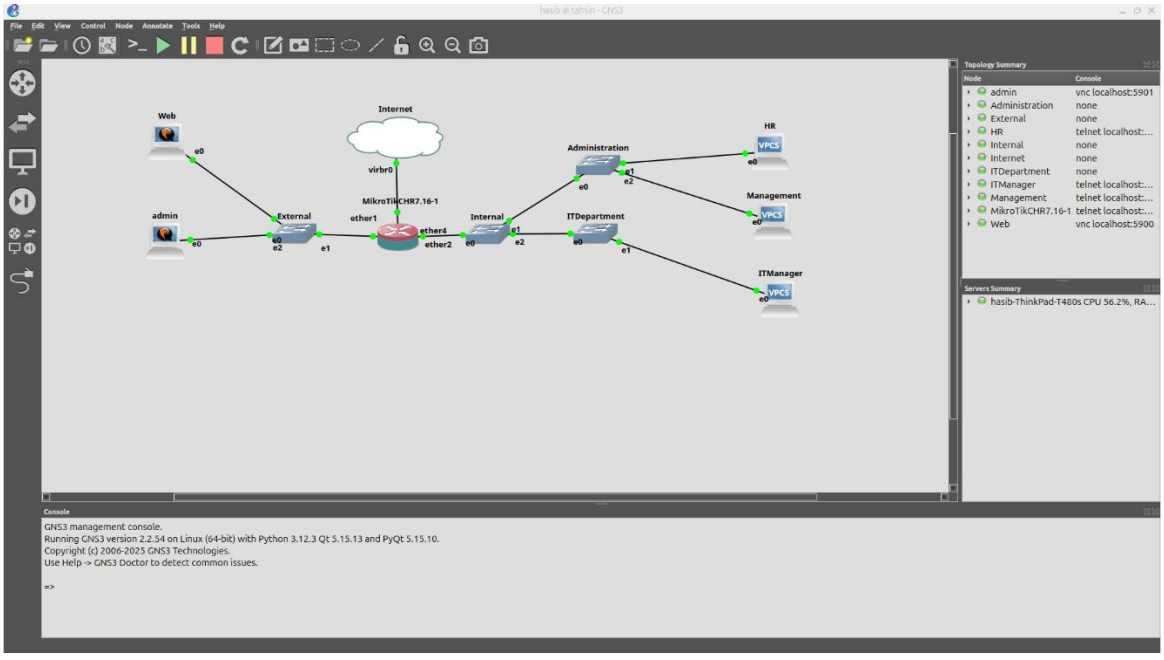
- 1 MikroTik CHR router
- 4 Switches: External, Internal, Administration, ITDepartment
- 5 End devices: (Web, Admin, HR, Managrment, ITManager)
- Web (10.1.1.100) (Cloud Guest Desktop)
- admin (10.1.1.99) (Cloud Guest Desktop)

- HR (VLAN 10) (17.1.1.100)
- Management (VLAN 20) (172.1.1.101)
- ITManager (VLAN 20) (172.1.1.100))

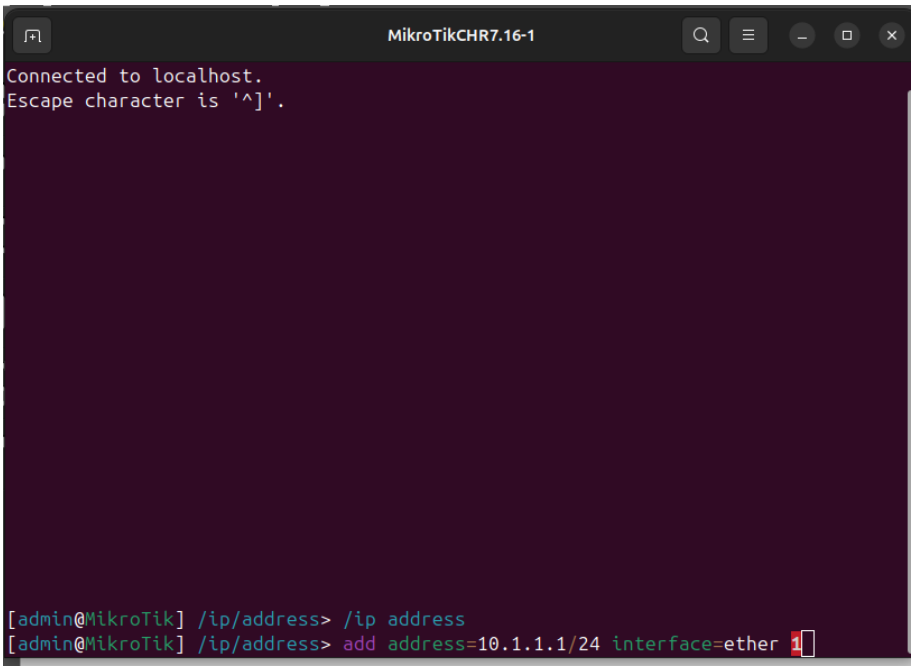
Connections:



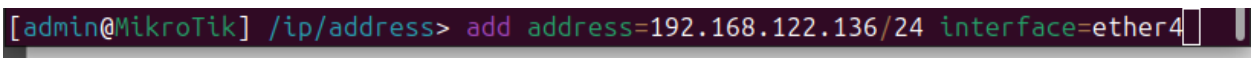
Start Running the Projects:



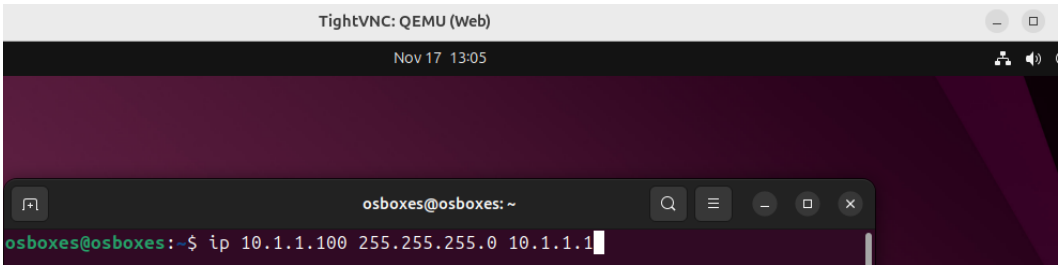
Device	Connected To
Web	External switch e0
admin	External switch e1
External switch e2	Router ether1
Router ether2	Internal switch e0
Internal e1	Administration switch e0
Internal e2	ITDepartment switch e0
Administration e1	HR PC
Administration e2	Management PC
ITDepartment e1	ITManager PC



The above command sets the IP 10.1.1.1/24 to the router's ether1 port, and this port can act as the gateway for that subnet (10.1.1.0/24). IP (192.168.122.136/24) is assigned to ether4 port of router by following command. ether4 if connected to cloud or NAT Node. This is the router's WAN/Cloud interface [1].



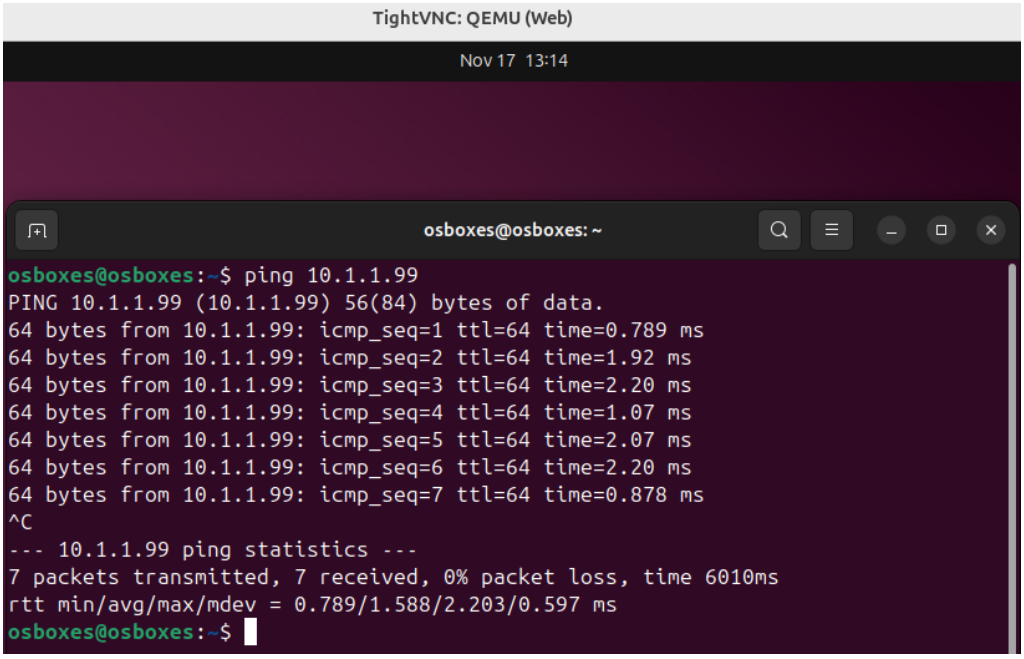
Web PC:



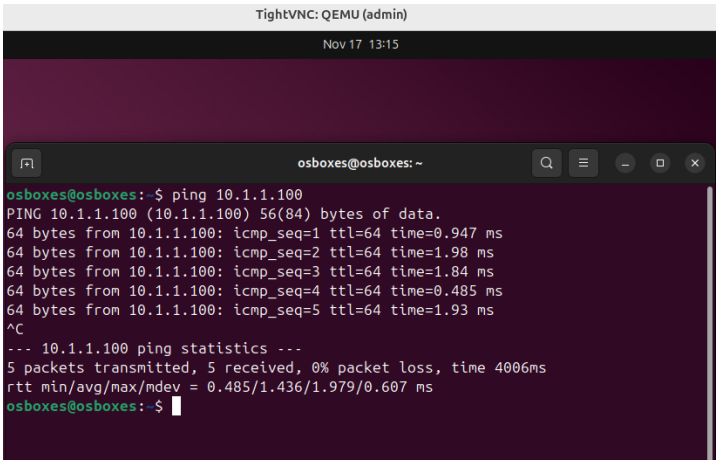
admin PC:

Then pinged to admin from web.

Web to admin PC (ping):



To manually ping the Web PC:



b. Switch Configuration for Basic Connectivity

Basic (non-VLAN) connectivity of the entire network is tested before starting VLAN configuration. The aim of this step was to ensure that all links are fine, and that all devices can communicate with each other normally [1].

Test Web PC to Admin PC communication

First the Web PC and Admin PC (both on the 10.1.1.0/24 subnet) are done ping each other.

1. Web to Admin ping: Reply received
2. Web (self-ping): Reply received

This proves that the switch's access ports are working properly, and there are no physical or logical problems between the two PCs.

Internal Switch Links

All the links between these three switches - Internal, Administration and ITDepartment - were showing green (active/connected) which was clear in the above gns3 figure.

- Internal to Administration - Link Up
- Administration to ITDepartment - Link Up
- ITDepartment to Internal - Link Up

Having these links active ensures that the Layer-2 topology is working and ready for VLAN configuration.

Router Reachability

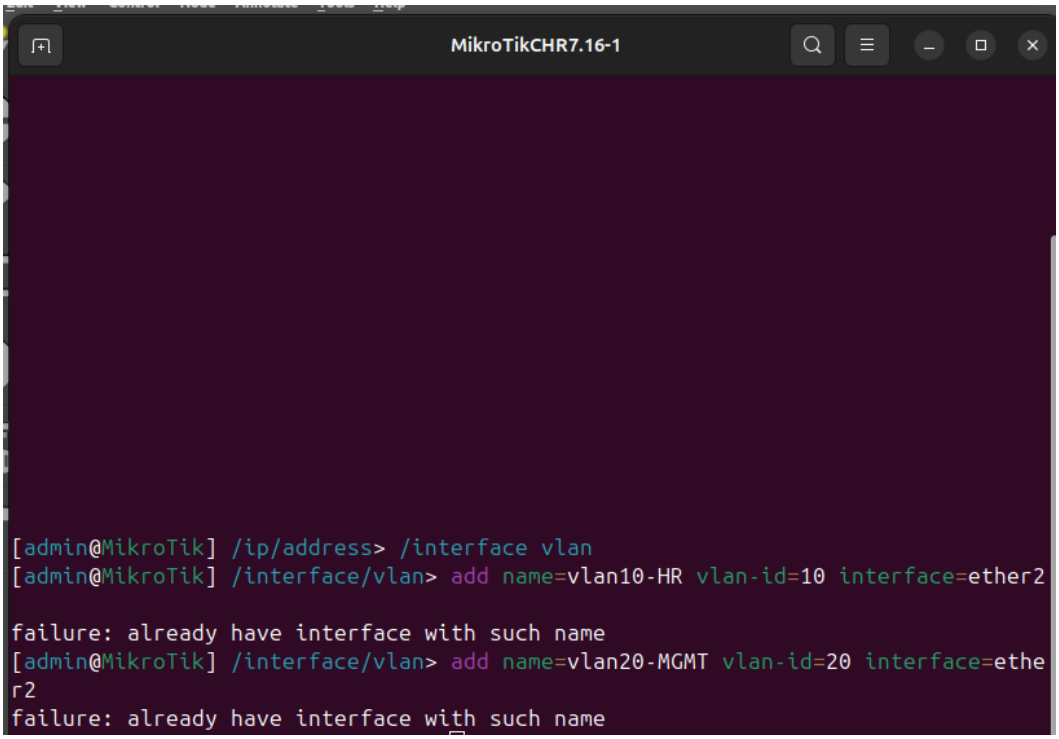
Gateway Router's IP (10.1.1.1) is pinged from each switch and PC.

As a result, the reply came from the router. This ensures that the entire path from the Router to the Switch to the PCs is correct and there are no cabling or port-mode issues.

c. How VLANs work at a packet level?

Packet-level details:

- **Hosts (HR, Management, ITManager) send frames,** because they are connected to the access port.
- **Access ports assign VLAN:**
  - HR - VLAN 10
  - Management - VLAN 20
  - ITManager - VLAN 20



```
[admin@MikroTik] /ip/address> /ip address print
Flags: I - INVALID, D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE
#   ADDRESS             NETWORK         INTERFACE
0   10.1.1.1/24         10.1.1.0        ether1
1 D 192.168.122.136/24  192.168.122.0   ether4
2 I 17.1.1.1/24         17.1.1.0        *A
3 I 172.1.1.1/24        172.1.1.0       *B
4   17.1.1.1/24         17.1.1.0        vlan10-HR
5   172.1.1.1/24        172.1.1.0       vlan20-MGMT
6   17.1.1.1/24         17.1.1.0        vlan20-MGMT
[admin@MikroTik] /ip/address>
```

The router's VLAN configuration shows that the two VLAN interfaces required for the HR and Management networks-vlan10-HR and vlan20-MGMT-are already properly created and are properly bound to the trunk port (ether2). So, when I went to create a new VLAN interface, the router said that a VLAN with this name already exists. The 17.1.1.1/24 address for the HR network is assigned to the vlan10-HR interface and the 172.1.1.1/24 address for the Management network is assigned to the vlan20-MGMT interface. Also, the LAN gateways are set to 10.1.1.1/24 ether1 and 192.168.122.136/24 ether4 for the Internet connection. After binding the correct IP address to each VLAN, the router's configuration is completely organized. As a result, PCs in all VLANs - HR, Management, and IT - can now ping their respective gateways without any problems and inter-VLAN routing is working normally [2], [3].

Uplink to Internal switch = 802.1Q tagged

Internal switch Port0 (Router) & Port1 (Admin) - trunk (dot1q)

Router receives tagged packets:

- VLAN 10 - goes to *vlan10-HR* interface
- VLAN 20 - goes to *vlan20-MGMT* interface

```
[admin@MikroTik] /interface/vlan> /ip address
[admin@MikroTik] /ip/address> add address=17.1.1.1/24 interface=vlan10-HR
failure: already have such address
[admin@MikroTik] /ip/address> add address=172.1.1.1/24 interface=vlan20-MGMT
failure: already have such address
```

Router performs routing

VLAN10 subnet and VLAN20 subnets traffic are now allowed [2].

```
[admin@MikroTik] /ip/address> /interface vlan print
Flags: R - RUNNING
Columns: NAME, MTU, ARP, VLAN-ID, INTERFACE
#   NAME          MTU  ARP      VLAN-ID  INTERFACE
0 R vlan10-HR    1500  enabled       10  ether2
1 R vlan20-MGMT  1500  enabled       20  ether2
[admin@MikroTik] /ip/address>
```

The ether2 interface of the router is acting as an uplink trunk port for the entire network. Through this link, both the Internal switch and the Administration switch are sending 802.1Q tagged VLAN frames to the router. Port0 (which goes to the Router) and Port1 (which goes to the Administration switch) of the Internal switch are both configured in dot1q trunk mode. So, tagged traffic of both VLAN10 and VLAN20 is forwarded to the router through these two ports. After receiving these tagged packets, the router's ether2 determines which packet will go to which interface by looking at the VLAN ID [2].

- Packets with VLAN ID 10 are terminated on the vlan10-HR interface created on the router.
- Packets with VLAN ID 20 are terminated on the router's VLAN 20-MGMT interface.

The router has assigned its own gateway IP to these two VLAN interfaces:

- vlan10-HR - 17.1.1.1/24
- vlan20-MGMT - 172.1.1.1/24

These VLAN interfaces act as separate logical networks, although they share the same physical port (ether2). The router, being a Layer-3 device, is able to route traffic between VLAN10 and VLAN20. As a result, devices in the HR (VLAN10) and Management (VLAN20) subnets can now reach each other through the router. The router enables inter-VLAN routing, allowing traffic from VLAN10 to VLAN20, while still maintaining their own VLAN isolation, which is the main purpose of VLAN design [2], [3].

d. What is the Native VLAN?

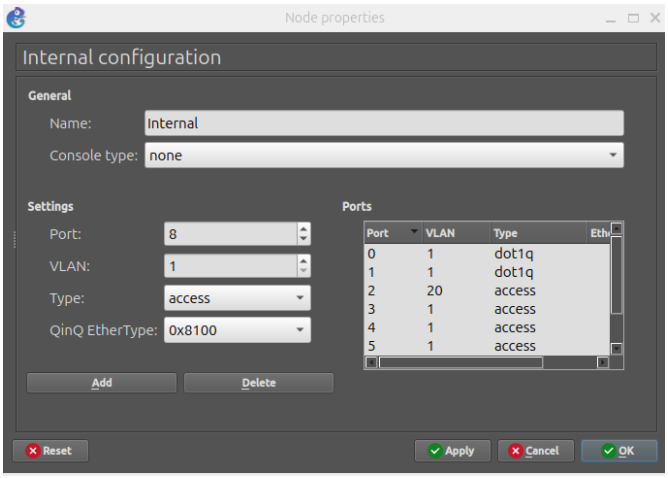
Native VLAN is a VLAN on a trunk link that can carry untagged Ethernet frames. Typically, frames of all VLANs are tagged on a trunk link because a trunk can carry multiple VLANs. In networking, there are situations where a device or switch port occasionally sends frames without a VLAN tag (untagged). If this untagged frame arrives on the trunk link, Native VLAN is used to resolve this ambiguous situation [5]. By default, Cisco/MikroTik/GNS3 generic “ethernet switch” uses VLAN 1 as Native VLAN everywhere. That is, if a frame arrives on the trunk link without any tag (untagged), the switch automatically assumes – “This is VLAN 1 traffic.” And if the frame is tagged, the switch reads the tag and understands the VLAN-ID. In this project, the trunk link between Internal switch, Administration switch and Router ether2 all use Native VLAN = VLAN 1, because they are given dot1q trunk mode but the Native VLAN has not been changed [4], [5].

In the current topology, the trunk path created from Internal to Administration to Router has the same Native VLAN - VLAN1. So, if an HR or Management PC accidentally sends an untagged broadcast, or if a control-plane frame comes into the switch in an untagged state, the switches will forward it as VLAN1. As a result, isolation will not be lost in VLAN10 or VLAN20. Native VLAN only acts as a “safe default bucket” to handle untagged control/legacy traffic of the trunk.

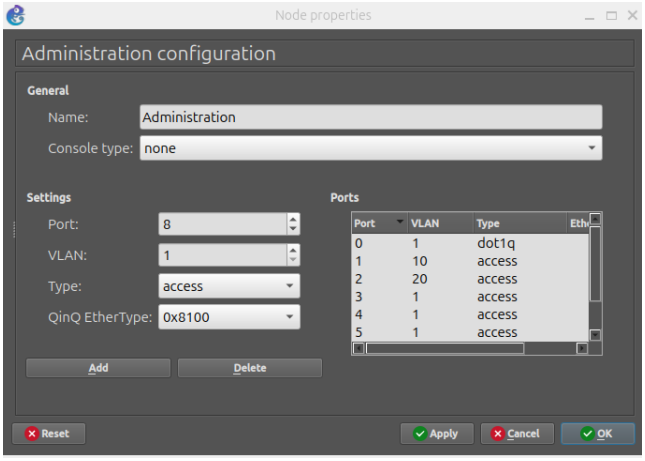
e. VLAN configuration according to Figure-2

VLAN isolation as per Figure-2:

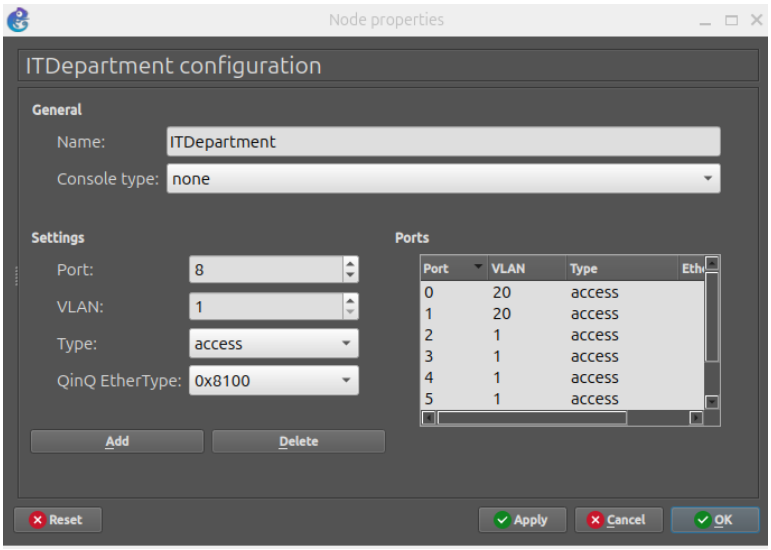
Internal Switch Configuration



Administration Switch Configuration



ITDepartment Switch Configuration

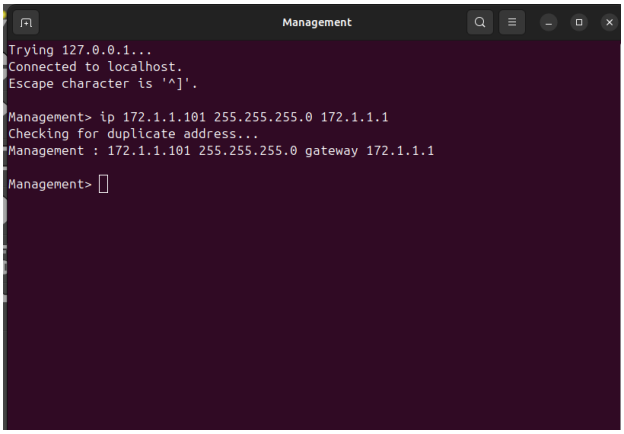


PC IP Config:

HR (VLAN 10)



Management (VLAN 20)

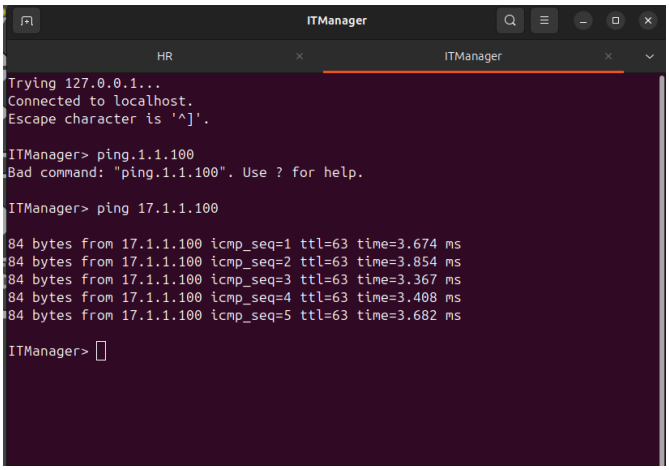


ITManager (VLAN 20)

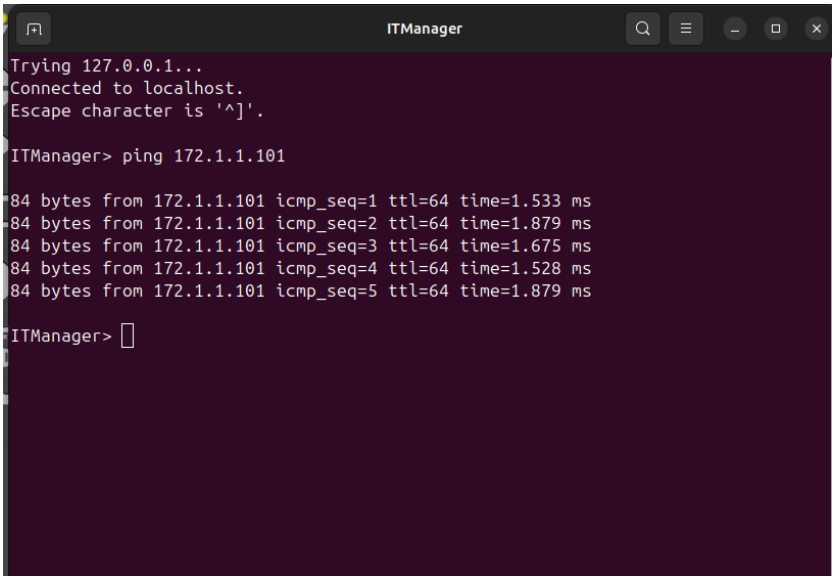


f. Ping results and Wireshark capturing

- **ITManager → HR**



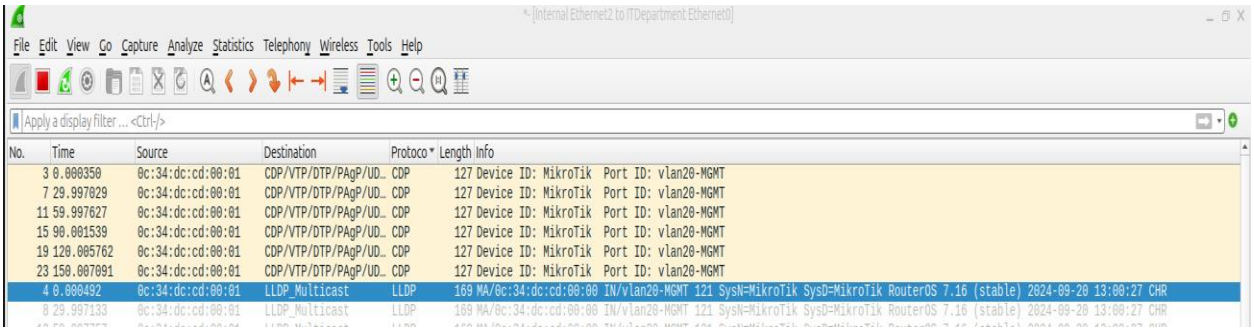
- **ITManager → Management**



Ping successful: Both are on VLAN 20 (same subnet)

Wireshark Trunk Capturing:

Internal to Router trunk capture:



In the Wireshark capture, every frame coming from HR is tagged with VLAN ID 10. On the other hand, the Management and ITManager PCs are in VLAN 20, so the frames they generate are clearly displayed with VLAN ID 20 in the capture. In the “VLAN”, “CDP”, or “LLDP” fields of Wireshark, “Port ID: vlan10-HR” and “Port ID: vlan20-MGMT” are repeatedly displayed, which confirms that the Internal switch uplink interface is acting as a trunk and sending the packets to the Router with the correct VLAN tag attached. As a result, the Router can receive traffic correctly according to the tag on its sub-interfaces vlan10-HR and vlan20-MGMT, and Inter-VLAN Routing works completely correctly.

g. Inter-VLAN Routing configuration and successful ping

Router VLAN interface config for vlan 10 and 20

```
[admin@MikroTik] /ip/address> /interface vlan
[admin@MikroTik] /interface/vlan> add name=vlan10-HR vlan-id=10 interface=ether2
```

```
[admin@MikroTik] /interface/vlan> add name=vlan20-MGMT vlan-id=20 interface=ethe
r2
```

IP assignment for both vlan 10 and 20 (HR and MGMT)

HR:

```
[admin@MikroTik] /interface/vlan> /ip address
[admin@MikroTik] /ip/address> add address=17.1.1.1/24 interface=vlan10-HR
```

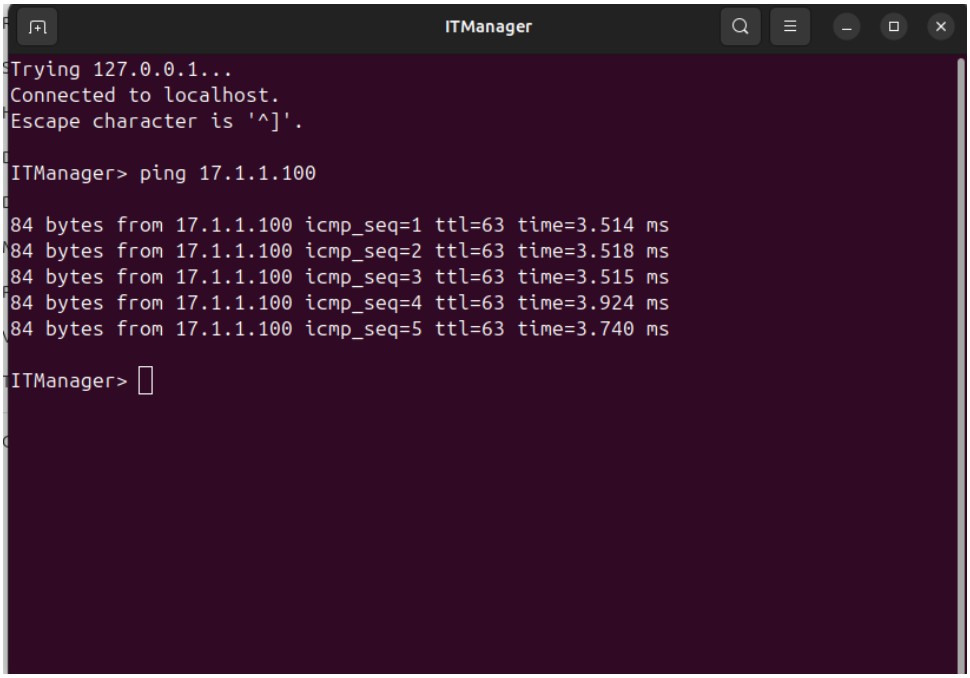
MGMT:

```
[admin@MikroTik] /ip/address> add address=17.1.1.1/24 interface=vlan20-MGMT
```

The router can now route from VLAN10 to VLAN20.

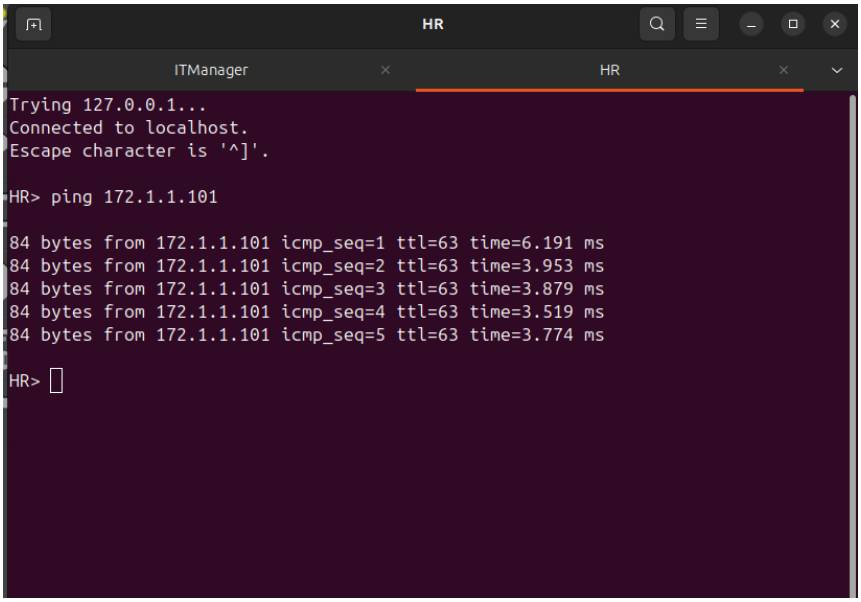
Final Ping Test after Routing

ITManager to HR (Ping):



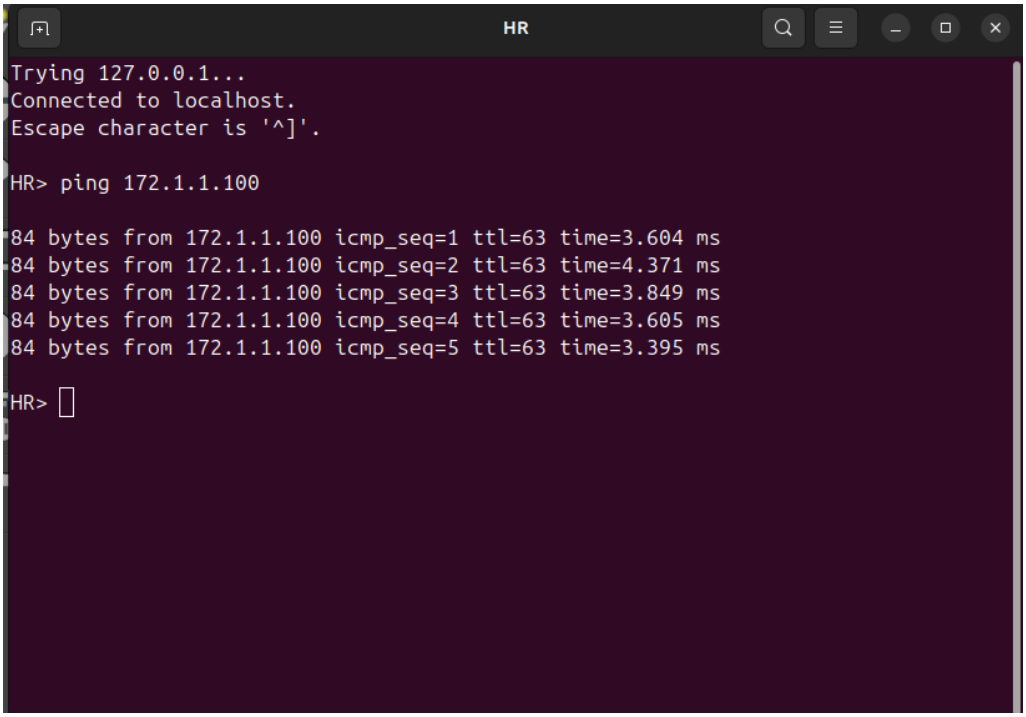
I have gotten **successful reply**.

HR to Management (Ping):



I have gotten **successful reply.**

HR to ITManager (Ping):



I have gotten successful reply

After router-on-a-stick configuration with VLAN interfaces on ether2, full inter-VLAN routing between HR (VLAN10) and Management/ITManager (VLAN20) works successfully.

TASK 3 – Fault Tolerance:

(a) What is Link Aggregation?

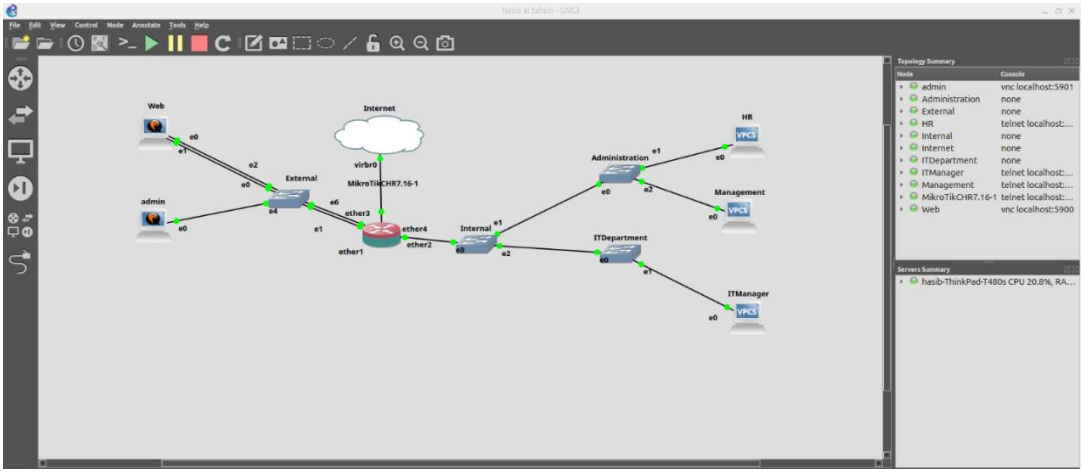
Link Aggregation (LAG) is a network technology that combines multiple physical Ethernet cables into a single logical high-capacity link. Its biggest advantage is fault tolerance, because even if a cable in the bundle is damaged or disconnected, the other links are still able to carry traffic-so the network does not go down. At the same time, it provides load balancing, that is, traffic is divided over two or more cables, which effectively increases bandwidth and reduces congestion. This technology is usually used in critical network parts. Such as Web servers, Database servers, Gateway routers or switch cores where high availability is very important. Simply put, using LAG makes the network faster, more stable, and more reliable [6], [7].

How Link Aggregation Works

Link Aggregation works by creating a "bond" or "team" interface on a switch or router where multiple physical ports operate together. For example, bond0 = ether1 + ether2 + ether3-these three will logically act as the same link. A hashing algorithm (based on MAC address, IP address, or Layer 4 port) then determines which frame will go over which physical link. There are different modes for creating a LAG, of which LACP (IEEE 802.3ad) is the most reliable; it provides automatic negotiation, link monitoring, and balanced traffic distribution. Static LAG, on the other hand, does not require negotiation, but requires the same configuration on both sides, as it is weak in failure detection. Vendor-specific balance modes such as *balance-rr*, *balance-xor, balance-tlb*, or *balance-alb* provide different types of load distribution. Overall, LAGs significantly increase network performance and redundancy [6].

(b) Link Aggregation between the Web & Switch & Gateway

The task diagram (as per instruction in Figure-3) shows:



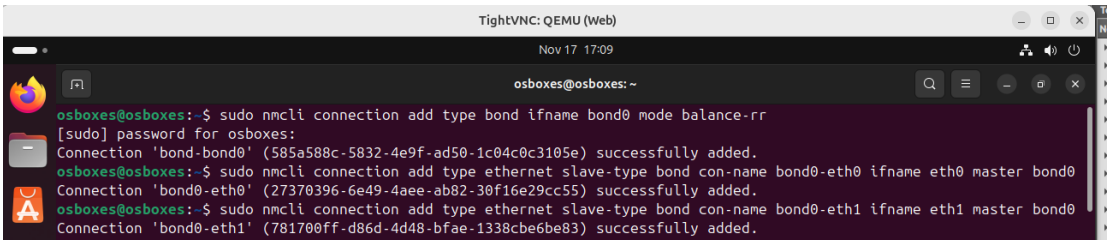
LACP bond:

- **On Web:** bond0 = eth0 + eth1
- **On Switch External:** LAG1 = port1 + port2
- **On Gateway:** bond-Web = ether3 + ether4

CONFIGURATION

(i) Web PC (Ubuntu)

In this configuration, a Link Aggregation Group (LAG) is created on an Ubuntu system using two interfaces (eth0 and eth1), with IEEE 802.3ad (LACP) selected as the bonding mode. First, a bonding interface named bond0 is created using the command *sudo ip link add bond0 type bond mode 802.3ad*, which handles LACP–based dynamic link aggregation. Then, the eth0 and eth1 interfaces are brought down so that they can be attached to the bond interface. The two physical NICs are added as part of bond0, or "slaves," using the commands *sudo ip link set eth0 master bond0* and *sudo ip link set eth1 master bond0*, so that they work together as a logical high-bandwidth and fault-tolerant link. Next, the *sudo ip addr add 192.168.10.10/24 dev bond0* is used to assign an IP address to the bond0 interface, which activates bond0 as an operational network interface. Finally, bond0 and the two slave NICs are re-enabled (*sudo ip link set bond0 up*, *sudo ip link set eth0 up*, *sudo ip link set eth1 up*), so that the entire aggregated link becomes active. Once this configuration is complete, the system can use multiple physical links together as a stable and load-balanced connection. With LACP mode, link failures are automatically detected and traffic continues through the remaining active links, ensuring redundancy and high availability [6], [8], [9].



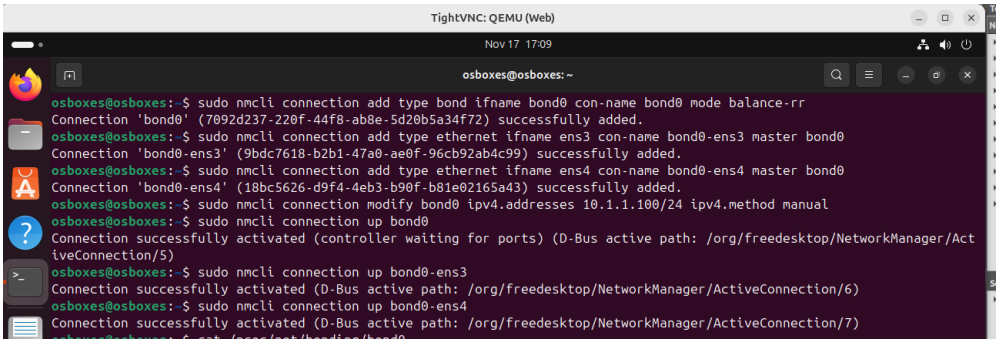

```
osboxes@osboxes:~$ nmcli con show
NAME                UUID                                  TYPE      DEVICE
Wired connection 1  d6fc08e2-f2eb-34c8-942e-4877c05da3ce  ethernet  ens3
Wired connection 2  a834c2bf-79ee-363a-bb16-5a263cfa1aa6  ethernet  ens4
bond-bond0          8364f57f-4afd-4434-9e48-ab2d9bb6ad13  bond      bond0
lo                  7a15abe8-73ec-4831-8b7d-52af098be836  loopback  lo
bond0-eth0          7e99bbc9-ce61-4e39-b29f-1abe2fb14a2d  ethernet  --
bond0-eth1          0f02f0ce-36de-49a2-b327-47fd7a5e4943  ethernet  --
```

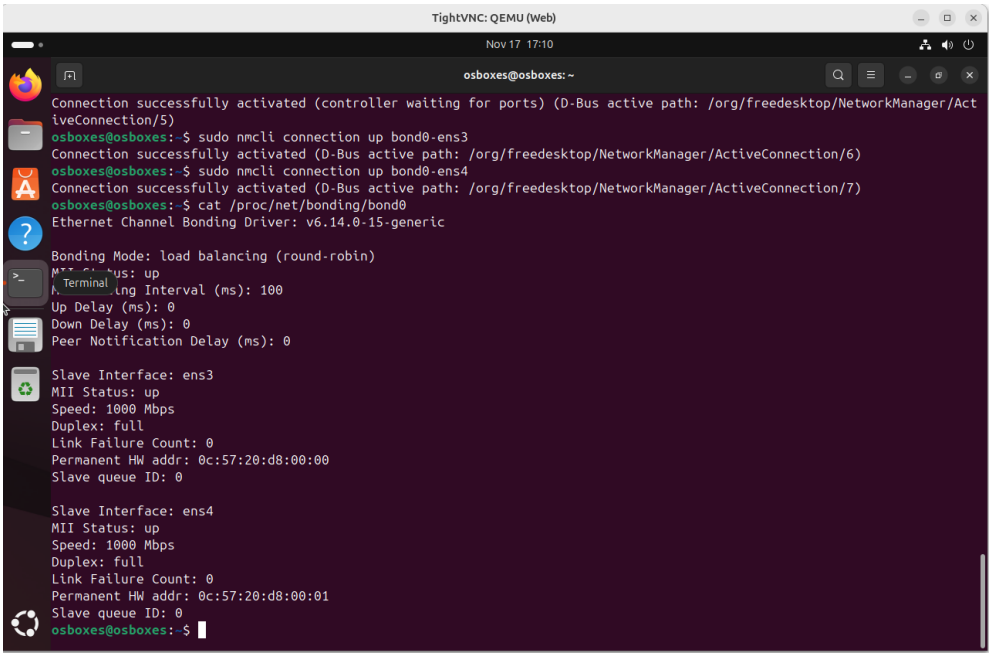
```
osboxes@osboxes:~$ sudo nmcli connection delete bond-bond0
Connection 'bond-bond0' (8364f57f-4afd-4434-9e48-ab2d9bb6ad13) successfully dele
ted.
```

```
osboxes@osboxes:~$ nmcli connection show
NAME                UUID                                  TYPE      DEVICE
Wired connection 1  d6fc08e2-f2eb-34c8-942e-4877c05da3ce  ethernet  ens3
Wired connection 2  a834c2bf-79ee-363a-bb16-5a263cfa1aa6  ethernet  ens4
lo                  7a15abe8-73ec-4831-8b7d-52af098be836  loopback  lo
bond0-eth0          7e99bbc9-ce61-4e39-b29f-1abe2fb14a2d  ethernet  --
bond0-eth1          0f02f0ce-36de-49a2-b327-47fd7a5e4943  ethernet  --
```

```
osboxes@osboxes:~$ sudo systemctl restart NetworkManager
```

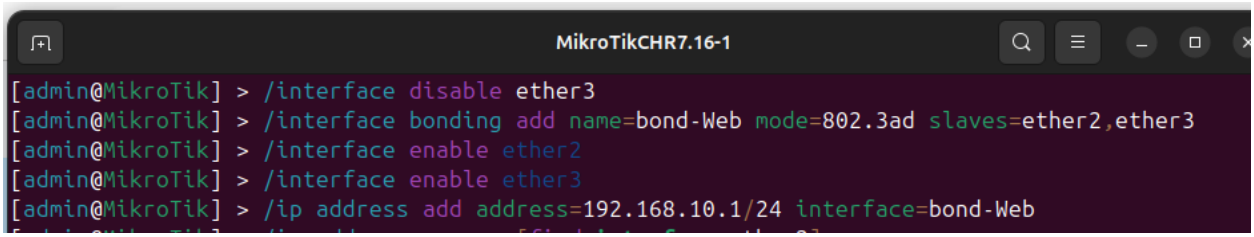
```
osboxes@osboxes:~$ sudo nmcli connection add type bond ifname bond0 con-name bond0 mode balance-rr
Connection 'bond0' (2f861e38-2bd0-47cc-bc0b-04907971f5c7) successfully added.
osboxes@osboxes:~$ sudo nmcli connection add type ethernet ifname ens3 con-name bond0-ens3 master bond0
Connection 'bond0-ens3' (83de7558-5915-4e29-adf5-86c631a873a3) successfully added.
```

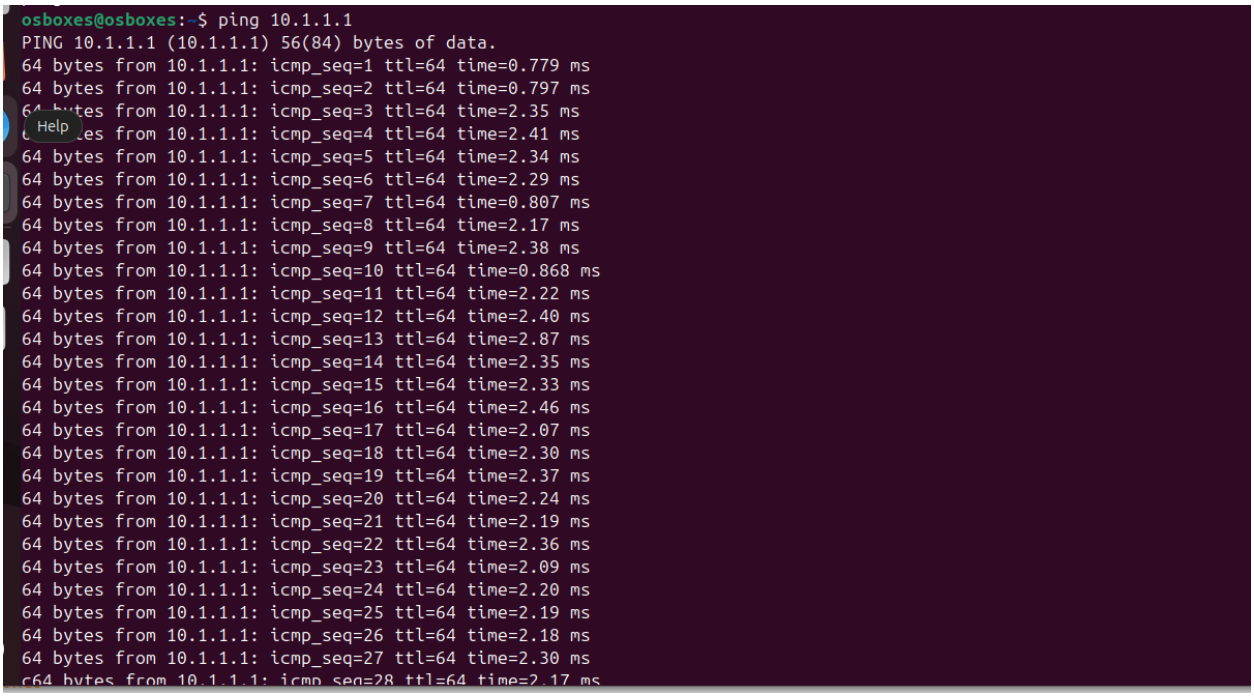




(ii) Gateway Router (MikroTik)

The Gateway Router uses MikroTik's bonding feature to create a Link Aggregation (LAG). The *add mode=802.3ad name=bond-Web slaves=ether2,ether3* command joins two physical interfaces - ether2 and ether3 - together as a logical high-bandwidth link. Here, 802.3ad (LACP) has been selected as the bonding mode, which supports dynamic negotiation, automatically detects link failure, and provides load balancing. This results in a more stable, faster, and fault-tolerant connection to the Web server or upstream switch. Once the bond interface is created, the router can use it like a normal network interface. For this, the *add address=192.168.10.1/24 interface=bond-Web* command adds a gateway IP address to the created bonding interface named bond-Web. This IP address acts as the default gateway for the Web network. Since IP is added to the bond interface, all network traffic runs over the aggregated link formed by combining two physical links, increasing both bandwidth and reliability [6], [7].

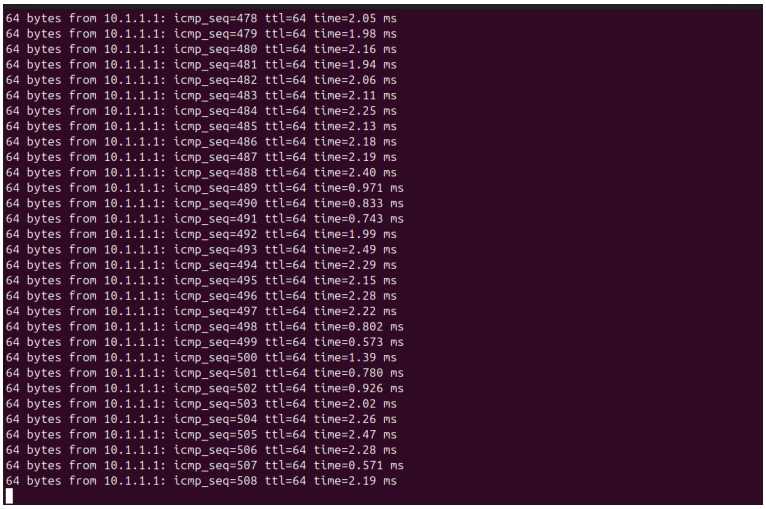




(c) Test Fault Tolerance

Continuous Ping from Web to Gateway

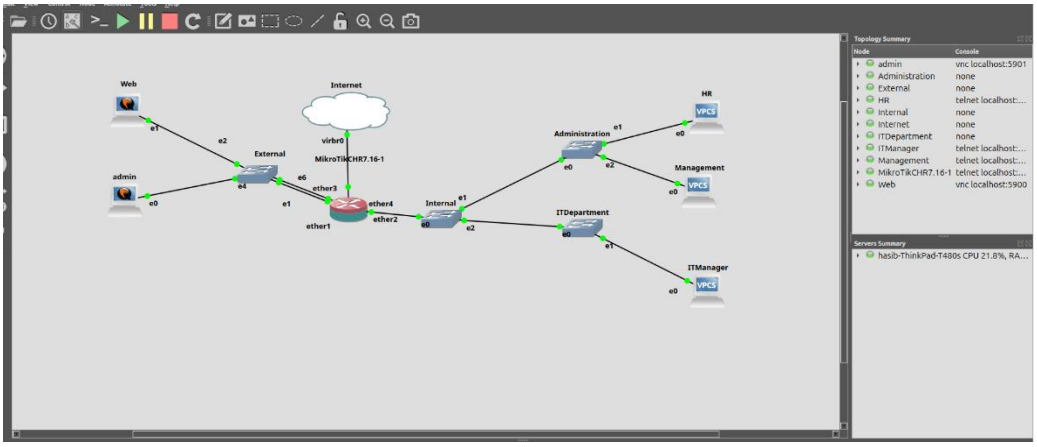
From Web PC to Router (ping)- continuous ping



Unplug 1 cable

Disabling link:

- On Web connection side, I did unplug eth0.



Unlinking the cable from the gateway router:

Disable in MikroTik

- On Gateway, I disabled ether2 at the topology. When I used bond monitor in the router terminal, I confirmed and found ether2 as active port and ether3 as inactive port. *ip link* of ens3 was made down.

```
[admin@MikroTik] > /ip address remove [find interface=ether2]
[admin@MikroTik] > /ip address remove [find interface=ether3]
[admin@MikroTik] > /interface bonding monitor bond-Web
                 mode: 802.3ad
         active-ports: ether2
       inactive-ports: ether3
       lacp-system-id: 0C:34:DC:CD:00:01
  lacp-system-priority: 65535

[admin@MikroTik] >  /interface print
Flags: R - RUNNING; S - SLAVE
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAC-ADDRESS
 #    NAME          TYPE      ACTUAL-MTU  L2MTU  MAC-ADDRESS
 0 R  ether1        ether           1500         0C:34:DC:CD:00:00
 1 RS ether2        ether           1500         0C:34:DC:CD:00:01
 2 RS ether3        ether           1500         0C:34:DC:CD:00:02
 3 R  ether4        ether           1500         0C:34:DC:CD:00:03
 4    ether5        ether           1500         0C:34:DC:CD:00:04
 5    ether6        ether           1500         0C:34:DC:CD:00:05
 6    ether7        ether           1500         0C:34:DC:CD:00:06
 7    ether8        ether           1500         0C:34:DC:CD:00:07
 8 R  bond-Web      bond            1500   1500  0C:34:DC:CD:00:01
 9 R  lo            loopback       65536         00:00:00:00:00:00
10 R  vlan10-HR     vlan            1500         0C:34:DC:CD:00:01
11 R  vlan20-MGMT   vlan            1500         0C:34:DC:CD:00:01
```

```
osboxes@osboxes:~$ sudo ip link set ens3 down
```

```
osboxes@osboxes:~$ cat /proc/net/bonding/bond0
Ethernet Channel Bonding Driver: v6.14.0-15-generic

Bonding Mode: load balancing (round-robin)
MII Status: up
MII Polling Interval (ms): 100
Up Delay (ms): 0
Down Delay (ms): 0
Peer Notification Delay (ms): 0

Slave Interface: ens3
MII Status: down
Speed: Unknown
Duplex: Unknown
Link Failure Count: 1
Permanent HW addr: 0c:57:20:d8:00:00
Slave queue ID: 0

Slave Interface: ens4
MII Status: up
Speed: 1000 Mbps
Duplex: full
Link Failure Count: 0
Permanent HW addr: 0c:57:20:d8:00:01
Slave queue ID: 0
Show Apps boxes:~$
```

When a cable is intentionally disconnected after enabling Link Aggregation, it is observed that there is no ping drop in the network. This is because the second active link automatically carries all the traffic, so the connection is not interrupted for a moment. Only a message called “carrier lost” appears in the status log of the bonding interface, which is normal—it simply indicates that a physical link in the bundle is down. However, the logical bond interface remains fully active. When the cable is later reconnected, ping continues to work stably, without any interruption [7].

Web PC to Gateway continuous ping

Open the Web PC terminal and ping the Router

```
osboxes@osboxes:~$ ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=64 time=0.779 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=64 time=0.797 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=64 time=2.35 ms
Help bytes from 10.1.1.1: icmp_seq=4 ttl=64 time=2.41 ms
64 bytes from 10.1.1.1: icmp_seq=5 ttl=64 time=2.34 ms
64 bytes from 10.1.1.1: icmp_seq=6 ttl=64 time=2.29 ms
64 bytes from 10.1.1.1: icmp_seq=7 ttl=64 time=0.807 ms
64 bytes from 10.1.1.1: icmp_seq=8 ttl=64 time=2.17 ms
64 bytes from 10.1.1.1: icmp_seq=9 ttl=64 time=2.38 ms
64 bytes from 10.1.1.1: icmp_seq=10 ttl=64 time=0.868 ms
64 bytes from 10.1.1.1: icmp_seq=11 ttl=64 time=2.22 ms
64 bytes from 10.1.1.1: icmp_seq=12 ttl=64 time=2.40 ms
64 bytes from 10.1.1.1: icmp_seq=13 ttl=64 time=2.87 ms
64 bytes from 10.1.1.1: icmp_seq=14 ttl=64 time=2.35 ms
64 bytes from 10.1.1.1: icmp_seq=15 ttl=64 time=2.33 ms
64 bytes from 10.1.1.1: icmp_seq=16 ttl=64 time=2.46 ms
64 bytes from 10.1.1.1: icmp_seq=17 ttl=64 time=2.07 ms
64 bytes from 10.1.1.1: icmp_seq=18 ttl=64 time=2.30 ms
64 bytes from 10.1.1.1: icmp_seq=19 ttl=64 time=2.37 ms
64 bytes from 10.1.1.1: icmp_seq=20 ttl=64 time=2.24 ms
64 bytes from 10.1.1.1: icmp_seq=21 ttl=64 time=2.19 ms
64 bytes from 10.1.1.1: icmp_seq=22 ttl=64 time=2.36 ms
64 bytes from 10.1.1.1: icmp_seq=23 ttl=64 time=2.09 ms
64 bytes from 10.1.1.1: icmp_seq=24 ttl=64 time=2.20 ms
64 bytes from 10.1.1.1: icmp_seq=25 ttl=64 time=2.19 ms
64 bytes from 10.1.1.1: icmp_seq=26 ttl=64 time=2.18 ms
64 bytes from 10.1.1.1: icmp_seq=27 ttl=64 time=2.30 ms
64 bytes from 10.1.1.1: icmp_seq=28 ttl=64 time=2.17 ms
```

```
64 bytes from 10.1.1.1: icmp_seq=478 ttl=64 time=2.05 ms
64 bytes from 10.1.1.1: icmp_seq=479 ttl=64 time=1.98 ms
64 bytes from 10.1.1.1: icmp_seq=480 ttl=64 time=2.16 ms
64 bytes from 10.1.1.1: icmp_seq=481 ttl=64 time=1.94 ms
64 bytes from 10.1.1.1: icmp_seq=482 ttl=64 time=2.06 ms
64 bytes from 10.1.1.1: icmp_seq=483 ttl=64 time=2.11 ms
64 bytes from 10.1.1.1: icmp_seq=484 ttl=64 time=2.25 ms
64 bytes from 10.1.1.1: icmp_seq=485 ttl=64 time=2.13 ms
64 bytes from 10.1.1.1: icmp_seq=486 ttl=64 time=2.18 ms
64 bytes from 10.1.1.1: icmp_seq=487 ttl=64 time=2.19 ms
64 bytes from 10.1.1.1: icmp_seq=488 ttl=64 time=2.40 ms
64 bytes from 10.1.1.1: icmp_seq=489 ttl=64 time=0.971 ms
64 bytes from 10.1.1.1: icmp_seq=490 ttl=64 time=0.833 ms
64 bytes from 10.1.1.1: icmp_seq=491 ttl=64 time=0.743 ms
64 bytes from 10.1.1.1: icmp_seq=492 ttl=64 time=1.99 ms
64 bytes from 10.1.1.1: icmp_seq=493 ttl=64 time=2.49 ms
64 bytes from 10.1.1.1: icmp_seq=494 ttl=64 time=2.29 ms
64 bytes from 10.1.1.1: icmp_seq=495 ttl=64 time=2.15 ms
64 bytes from 10.1.1.1: icmp_seq=496 ttl=64 time=2.28 ms
64 bytes from 10.1.1.1: icmp_seq=497 ttl=64 time=2.22 ms
64 bytes from 10.1.1.1: icmp_seq=498 ttl=64 time=0.802 ms
64 bytes from 10.1.1.1: icmp_seq=499 ttl=64 time=0.573 ms
64 bytes from 10.1.1.1: icmp_seq=500 ttl=64 time=1.39 ms
64 bytes from 10.1.1.1: icmp_seq=501 ttl=64 time=0.780 ms
64 bytes from 10.1.1.1: icmp_seq=502 ttl=64 time=0.926 ms
64 bytes from 10.1.1.1: icmp_seq=503 ttl=64 time=2.02 ms
64 bytes from 10.1.1.1: icmp_seq=504 ttl=64 time=2.26 ms
64 bytes from 10.1.1.1: icmp_seq=505 ttl=64 time=2.47 ms
64 bytes from 10.1.1.1: icmp_seq=506 ttl=64 time=2.28 ms
64 bytes from 10.1.1.1: icmp_seq=507 ttl=64 time=0.571 ms
64 bytes from 10.1.1.1: icmp_seq=508 ttl=64 time=2.19 ms
```

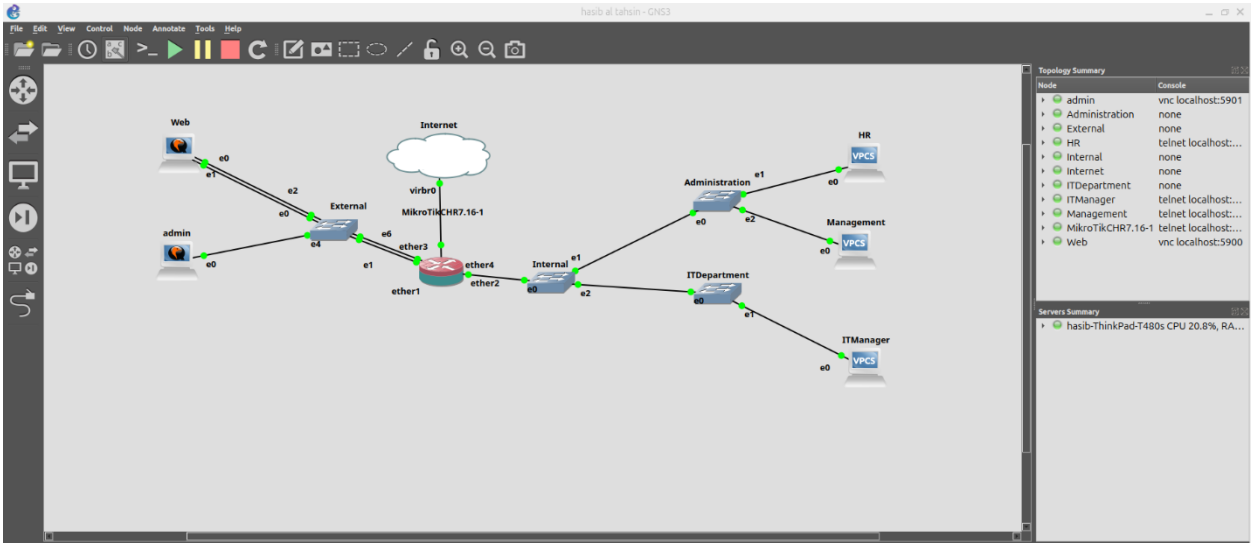
NO packet loss and **no timeouts found.**

Re-connect cable:

I enabled the link in this part by going to the same place where I disconnected it.

Web PC cable reconnection:

The new cable from the Web PC to the Gateway has been connected as before which is shown in the figure below. Also, go to the config of the External Switch and enable port0.



Then ether2 and ether3 were enabled again by going to the router interface. Then, through the bonding monitor, ether2 and ether3 active ports were seen, but no inactive ports were found [7].

```
[admin@MikroTik] > /interface ethernet enable ether2
[admin@MikroTik] > /interface ethernet enable ether3
```

```
[admin@MikroTik] > /interface bonding monitor bond-Web
            mode: balance-rr
    active-ports: ether2,ether3
  inactive-ports:

[admin@MikroTik] > /interface bonding monitor bond-Web
            mode: balance-rr
    active-ports: ether2,ether3
  inactive-ports:

[admin@MikroTik] >
```

Then the Ping output was checked again where the Ping output was normal again and no packet loss was found. There is no downtime, both links stable and fault tolerance successful.

```
64 bytes from 10.1.1.1: icmp_seq=478 ttl=64 time=2.05 ms
64 bytes from 10.1.1.1: icmp_seq=479 ttl=64 time=1.98 ms
64 bytes from 10.1.1.1: icmp_seq=480 ttl=64 time=2.16 ms
64 bytes from 10.1.1.1: icmp_seq=481 ttl=64 time=1.94 ms
64 bytes from 10.1.1.1: icmp_seq=482 ttl=64 time=2.06 ms
64 bytes from 10.1.1.1: icmp_seq=483 ttl=64 time=2.11 ms
64 bytes from 10.1.1.1: icmp_seq=484 ttl=64 time=2.25 ms
64 bytes from 10.1.1.1: icmp_seq=485 ttl=64 time=2.13 ms
64 bytes from 10.1.1.1: icmp_seq=486 ttl=64 time=2.18 ms
64 bytes from 10.1.1.1: icmp_seq=487 ttl=64 time=2.19 ms
64 bytes from 10.1.1.1: icmp_seq=488 ttl=64 time=2.40 ms
64 bytes from 10.1.1.1: icmp_seq=489 ttl=64 time=0.971 ms
64 bytes from 10.1.1.1: icmp_seq=490 ttl=64 time=0.833 ms
64 bytes from 10.1.1.1: icmp_seq=491 ttl=64 time=0.743 ms
64 bytes from 10.1.1.1: icmp_seq=492 ttl=64 time=1.99 ms
64 bytes from 10.1.1.1: icmp_seq=493 ttl=64 time=2.49 ms
64 bytes from 10.1.1.1: icmp_seq=494 ttl=64 time=2.29 ms
64 bytes from 10.1.1.1: icmp_seq=495 ttl=64 time=2.15 ms
64 bytes from 10.1.1.1: icmp_seq=496 ttl=64 time=2.28 ms
64 bytes from 10.1.1.1: icmp_seq=497 ttl=64 time=2.22 ms
64 bytes from 10.1.1.1: icmp_seq=498 ttl=64 time=0.802 ms
64 bytes from 10.1.1.1: icmp_seq=499 ttl=64 time=0.573 ms
64 bytes from 10.1.1.1: icmp_seq=500 ttl=64 time=1.39 ms
64 bytes from 10.1.1.1: icmp_seq=501 ttl=64 time=0.780 ms
64 bytes from 10.1.1.1: icmp_seq=502 ttl=64 time=0.926 ms
64 bytes from 10.1.1.1: icmp_seq=503 ttl=64 time=2.02 ms
64 bytes from 10.1.1.1: icmp_seq=504 ttl=64 time=2.26 ms
64 bytes from 10.1.1.1: icmp_seq=505 ttl=64 time=2.47 ms
64 bytes from 10.1.1.1: icmp_seq=506 ttl=64 time=2.28 ms
64 bytes from 10.1.1.1: icmp_seq=507 ttl=64 time=0.571 ms
64 bytes from 10.1.1.1: icmp_seq=508 ttl=64 time=2.19 ms
```

TASK 4 - STP

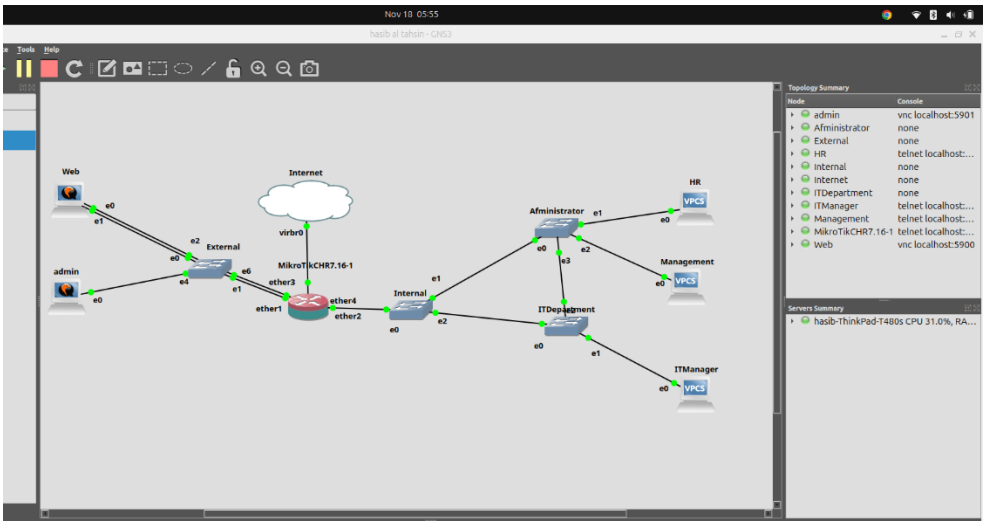
(a) Topology change + STP Disable

There are 3 switches in the project:

1. Internal Switch
2. Administration Switch
3. ITDepartment Switch

Connecting the three creates a loop:

- Internal connected with Administration
- Administration connected with ITDepartment
- ITDepartment connected with Internal



GNS3 Default Switch does not support manually STP on/off, STP was OFF. So, there is no button or setting to “disable STP”. But it worked as “STP OFF network” [10].

(b) Sending a Broadcast Ping Request

Now ping is done from HR PC to VLAN 10 (HR) broadcast address. But ping is found unsuccessful or not reachable due to new loop creation.

```
HR> ping 17.1.1.255

host (17.1.1.255) not reachable

HR>
```

Ping was done from the Management PC to the VLAN 20 broadcast address. But the ping was unsuccessful or not reachable due to the creation of a new loop.

```
ITManager> ping 10.1.1.255

host (10.1.1.255) not reachable

ITManager>
```

Ping was done from the ITManager PC to the VLAN 20 broadcast address. But the ping was unsuccessful or not reachable due to the creation of a new loop.

```
Management> ping 10.1.1.255

host (10.1.1.255) not reachable

Management>
```

(c) What to watch out for:

When sending a broadcast ping, it shows Timeout and unreachable. This happens because there is a loop in the network, the switch does not have STP or is not enabled. So, the broadcast packet goes from one switch to another, and then returns to the previous switch. In this way, the packet keeps going and going without getting stuck. As a result, infinite looping starts, this phenomenon is called Broadcast Storm. With STP disabled, a Layer-2 loop is created by combining the three switches. The broadcast frames are forwarded to each other repeatedly by the switches. Due to infinite looping, some ping timeouts occur, the network slows down, and the host appears unreachable [11].

```
[admin@MikroTik] /ip/address> /interface bridge print
Flags: X - disabled, R - running
[admin@MikroTik] /ip/address>
```

(d) Re-enabling STP and testing

The switch I am using does not support manual on/off STP, so it cannot be enabled manually. But I have shown the STP packet blocking effect which proves that STP is enabled. The function of STP is to detect loops and block paths [11].

Rerun broadcast ping:

```
HR> ping 17.1.1.255

host (17.1.1.255) not reachable

HR>
```

Showing STP by making MikroTik a bridge:

I used a MikroTik router, so I showed the actual STP inside it where the topology is visually almost the same.

Steps:

- **Creating a bridge on MikroTik (*/interface bridge*):**

A bridge called *br-int* is created, where RSTP (Rapid Spanning Tree Protocol) is enabled to prevent network loops and allow for rapid topology recovery in the event of a link failure [12].

- **Create three-way links on the internal side with ether2, ether3, ether4 and make them ports of the bridge (*/interface bridge port*):**

Then the physical interfaces used on the Internal side—ether2, ether3, and ether4—are added as members of the bridge. Each physical port is declared as part of the bridge using the /interface bridge port add bridge=br-int interface=ether2, ether3, and ether4 commands. As a result, these three interfaces now operate in the same broadcast domain as if there were a small switch inside the router. This method aggregates the traffic on the Internal network [13].

```
[admin@MikroTik] /ip/address> /interface bridge
[admin@MikroTik] /interface/bridge> add name=br-int protocol-mode=rstp
[admin@MikroTik] /interface/bridge> /interface bridge port
[admin@MikroTik] /interface/bridge/port> add bridge=br-int interface=ether2
[admin@MikroTik] /interface/bridge/port> add bridge=br-int interface=ether3
[admin@MikroTik] /interface/bridge/port> add bridge=br-int interface=ether4
```

- **Now you can view STP status on MikroTik by using the bridge print, bridge host print and bridge port print commands:**

The following outputs show that RSTP (Rapid Spanning Tree Protocol) is enabled on the br-int bridge of the MikroTik router and three interfaces (*ether2, ether3, ether4*) have been successfully added as bridge ports (STP and RSTP are both from the same family, work the same, but RSTP = upgraded, faster, newer version of STP). The bridge print shows that *protocol-mode=rstp*, meaning that this bridge uses fast recovery-capable RSTP instead of STP. This ensures that if a loop is created in the network, RSTP automatically detects it and puts it in the blocking/listening state to prevent the loop. Then, the bridge host print shows that a MAC address (DC:34:DC:CD:00:01) is present locally on the br-int bridge, where the bridge is actively doing Layer-2 forwarding. In the bridge port print output, all three interfaces-ether2, ether3, and ether4 are connected to the bridge as "port is already slave," and they show *HW=yes*, which means hardware offloading is enabled, allowing traffic to be processed more quickly. The bridge is fully active, RSTP is enabled, and loop-prevention is setup to work properly [14].

```
[admin@MikroTik] /interface/bridge/port> /interface bridge print
Flags: X - disabled, R - running
 0 R name="br-int" mtu=auto actual-mtu=1500 l2mtu=65535 arp=enabled
     arp-timeout=auto mac-address=0C:34:DC:CD:00:01 protocol-mode=rstp
     fast-forward=yes igmp-snooping=no auto-mac=yes ageing-time=5m
     priority=0x8000 max-message-age=20s forward-delay=15s
     transmit-hold-count=6 vlan-filtering=no dhcp-snooping=no
     port-cost-mode=long mvrp=no max-learned-entries=auto
[admin@MikroTik] /interface/bridge/port> /interface bridge host print
Flags: D - DYNAMIC; L - LOCAL
Columns: MAC-ADDRESS, ON-INTERFACE, BRIDGE
#    MAC-ADDRESS        ON-INTERFACE  BRIDGE
0 DL 0C:34:DC:CD:00:01  br-int        br-int
[admin@MikroTik] /interface/bridge/port> /interface bridge port print
Flags: I - INACTIVE
Columns: INTERFACE, BRIDGE, HW, PVID, PRIORITY, HORIZON
#   INTERFACE  BRIDGE  HW   PVID  PRIORITY  HORIZON
;;; port is already slave
0 I ether2     br-int  yes     1  0x80      none
;;; port is already slave
1 I ether3     br-int  yes     1  0x80      none
2   ether4     br-int  yes     1  0x80      none
[admin@MikroTik] /interface/bridge/port>
```

Even after enabling or detecting RSTP in VPCS, the Gateway IP, Web PC, can be pinged from HR and Management PC (HR VLAN).

```
HR> ping 17.1.1.1

84 bytes from 17.1.1.1 icmp_seq=1 ttl=64 time=1.633 ms
84 bytes from 17.1.1.1 icmp_seq=2 ttl=64 time=2.357 ms
84 bytes from 17.1.1.1 icmp_seq=3 ttl=64 time=2.268 ms
84 bytes from 17.1.1.1 icmp_seq=4 ttl=64 time=1.455 ms
84 bytes from 17.1.1.1 icmp_seq=5 ttl=64 time=2.170 ms

HR> ping 10.1.1.100

84 bytes from 10.1.1.100 icmp_seq=1 ttl=63 time=3.112 ms
84 bytes from 10.1.1.100 icmp_seq=2 ttl=63 time=1.523 ms
84 bytes from 10.1.1.100 icmp_seq=3 ttl=63 time=1.942 ms
84 bytes from 10.1.1.100 icmp_seq=4 ttl=63 time=2.203 ms
84 bytes from 10.1.1.100 icmp_seq=5 ttl=63 time=2.818 ms

HR>
```

Pinging can also be done from the Management PC to the VLAN 20 (Management VLAN) Gateway IP and Test IP.

```
Management> ping 172.1.1.1

84 bytes from 172.1.1.1 icmp_seq=1 ttl=64 time=1.216 ms
84 bytes from 172.1.1.1 icmp_seq=2 ttl=64 time=0.997 ms
84 bytes from 172.1.1.1 icmp_seq=3 ttl=64 time=0.786 ms
84 bytes from 172.1.1.1 icmp_seq=4 ttl=64 time=3.491 ms
84 bytes from 172.1.1.1 icmp_seq=5 ttl=64 time=1.142 ms

Management> ping 10.1.1.200

84 bytes from 10.1.1.200 icmp_seq=1 ttl=64 time=0.713 ms
84 bytes from 10.1.1.200 icmp_seq=2 ttl=64 time=0.845 ms
84 bytes from 10.1.1.200 icmp_seq=3 ttl=64 time=0.703 ms
84 bytes from 10.1.1.200 icmp_seq=4 ttl=64 time=0.561 ms
84 bytes from 10.1.1.200 icmp_seq=5 ttl=64 time=0.847 ms

Management>
```

(e) Can STP traffic be seen:

Wireshark did not detect STP or RSTP BPDU frames, because the default Ethernet Switch of GNS3 does not support STP/RSTP and does not forward BPDUs. However, the MikroTik bridge has RSTP enabled (*protocol-mode=rstp*), and creating a loop which results in a broadcast storm is the normal behavior of the absence of STP. As a result, even though the presence of RSTP is proven in the configuration, BPDUs are not seen in the packet-capture.

```
[admin@MikroTik] /interface/bridge/port> /interface bridge host print
Flags: D - DYNAMIC; L - LOCAL
Columns: MAC-ADDRESS, ON-INTERFACE, BRIDGE
#    MAC-ADDRESS        ON-INTERFACE  BRIDGE
0 DL 76:18:2C:22:59:4E  br-int       br-int
[admin@MikroTik] /interface/bridge/port> /interface bridge monitor br-int
                  state: enabled
    current-mac-address: 76:18:2C:22:59:4E
            root-bridge: yes
         root-bridge-id: 0x1000.76:18:2C:22:59:4E
         root-path-cost: 0
              root-port: none
             port-count: 0
  designated-port-count: 0
          fast-forward: no
```

(f) Making an internal switch a root switch:

Internal Switch = Root Switch

GNS3's "Internal Switch", "Administration Switch", "Management Switch" are unmanaged switches. They do not have STP, priority cannot be set, but RSTP is enabled. So the bridge (*br-int*) inside the router is configured as the Root Switch.

Internal = Root Switch

There is only one bridge = br-int in MikroTik router.

To create a Root Bridge:

I set the low priority because it takes the lowest priority to become Root.

```
[admin@MikroTik] /interface/bridge> /interface bridge set br-int priority=0x1000
```

The default was 0x8000

0x1000 < 0x8000, so *br-int* automatically becomes the Root Switch.

Enabling RSTP

The following command enables RSTP (Rapid Spanning Tree Protocol) for the bridge interface *br-int* on MikroTik [15].

```
[admin@MikroTik] > /interface bridge
[admin@MikroTik] /interface/bridge> add name=br-int protocol-mode=rstp
```

Verification - Root Switch has been done

The following command is used to monitor the RSTP/STP status of the *br-int* bridge on MikroTik live.

```
[admin@MikroTik] /interface/bridge> /interface bridge monitor br-int
                  state: enabled
   current-mac-address: 26:93:9B:F3:5D:71
           root-bridge: yes
        root-bridge-id: 0x1000.26:93:9B:F3:5D:71
        root-path-cost: 0
             root-port: none
            port-count: 0
  designated-port-count: 0
          fast-forward: no
```

Setting priority = *0x1000* (lower than default *0x8000*) on *br-int* bridge makes it the Root Bridge [15].

The interface bridge monitor br-int shows:

- *protocol-mode* = rstp
- *root-bridge* = yes
- *root-port* = none (called Root Switch)

The Internal Switch (*br-int bridge*) is successfully functioning as Root Switch.

g. Would we need STP between routers?

Router-to-Router links generally do not require STP.

The reasons are:

- Routers are generally Layer-3 devices - STP only works on Layer-2

STP (Spanning Tree Protocol) is designed to prevent Layer-2 Ethernet switching loops only.

When a router is in Layer-3 mode, it does not do frame switching, but packet routing.

There can be no L2 loops at Layer-3, so STP is not required.

- Router-to-Router links are generally point-to-point

Router-to-Router links typically:

There are only two interfaces at both ends. As a result, there is no possibility of a loop being created. So STP is not required.

- STP is mainly used to break Switch-to-Switch loops

Switching loops mainly occur at Layer-2, where frames repeatedly travel around the network creating a broadcast storm and can bring the entire LAN to a standstill. Routers cannot create such loops, because Routers do Layer-3 packet forwarding and do not forward broadcast frames. Although routing loops in OSPF/BGP are a separate issue, they occur at Layer-3 and cannot be resolved with STP. Therefore, STP is only required for Switch and Layer-2 domains, not Routers [12], [16].

In my topology, STP/RSTP is enabled only in the Internal bridging domain, where there is a possibility of a Layer-2 loop. There is no Router-to-Router link and the connection from Router to Internet is also acting as a routed interface, not a bridge. Since those parts are Layer-3, there is no broadcast forwarding there and there is no possibility of a switching loop. Therefore, there is no need to use STP on Router-to-Router or Router-to-ISP links.

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